

Seller Performance: Where Bad Customer Experiences Come From

A visual study of e-commerce orders, delivery and customer satisfaction.

Technical report: data preprocessing, exploratory analysis, and the two seller-performance questions.

Item	Detail
Dataset	Brazilian e-commerce marketplace, 9 relational tables
Period covered	September 2016 – August 2018
Orders analysed	94,563 (after cleaning)
Sellers analysed	2,944 (1,214 with 10+ orders)
Outcome measured	Rate of 1–2 star reviews ("bad experience")
Code file	<code>Seller_Performance_Analysis.ipynb</code>

1. Objective

The marketplace wants to grow without breaking the customer experience. Onboarding sellers aggressively lifts sales but risks late deliveries and one-star reviews. Cracking down on every slow seller protects satisfaction but shrinks the catalogue and stalls growth. These decisions are currently made on intuition.

This report answers the two seller-side questions that decide where the company should sit on that trade-off:

1. How concentrated is the damage across sellers? (the worst 5% of sellers)
2. Are certain sellers a recurring source of bad experiences regardless of category or delivery time?

Question 1 sizes the prize: if a few sellers cause most of the pain, a narrow intervention is enough; if not, no amount of seller policing will fix satisfaction. Question 2 decides whether intervention is fair — a seller that looks bad only because it sells a difficult category, or ships through a slow region, should not be penalised for it.

DEFINITION USED THROUGHOUT

A **"bad experience"** is a review of 1 or 2 stars out of 5. Across the cleaned data this is **12.35%** of orders (11,676 bad reviews).

We deliberately do not use the average star rating. The review distribution is U-shaped, so the mean of 4.1 sits in a gap where almost no customer actually is, and an average lets four 5-star orders cancel out a 1-star. A customer who has a bad experience churns; they do not average out.

2. Data preprocessing

Before any analysis we inspected the raw tables and found five properties that would have made a naive answer wrong. Each one forced a specific cleaning decision.

2.1 What we checked, and what we decided

#	What we found in the raw data	Decision taken
1	Only 97% of orders were delivered; the rest were cancelled, unavailable or still in transit. A review on an undelivered order measures a different problem.	Keep delivered orders only (96,478).
2	Reviews are recorded per ORDER, but sellers are attached to ITEMS. 1.3% of orders contain items from more than one seller, so one review cannot be assigned to one seller.	Keep single-seller orders only, so every review belongs to exactly one seller.
3	551 orders carried more than one review row, and the scores sometimes disagreed.	Keep the latest review per order — the customer's settled verdict.
4	8 orders marked delivered had no delivery timestamp, so no delay could be computed.	Drop them.
5	Half of all sellers have fewer than 9 orders. A seller with 3 orders and 1 bad review shows a 33% "bad rate" that means nothing.	Require 10+ orders before ranking any seller on a rate (1,214 sellers, 93.7% of orders).

2.2 How many orders survived each step

Cleaning step	Orders remaining
Raw orders table	99,441
Delivered only	96,478
Single-seller orders only	95,420
Has a review	94,571
Has a delivery timestamp — final analysis base	94,563

We kept 95.1% of all orders. Nothing was dropped for being inconvenient; each exclusion removes a record that cannot answer the question being asked.

ONE EXCLUSION IS WORTH REPORTING AS A FINDING

The 1,261 multi-seller orders we had to drop are **not** a random sample. They run a **47.3% bad-review rate** against 12.3% for single-seller orders.

Crucially, this is not caused by lateness: those orders are **less** likely to be late (1.3% vs 8.1%), and their bad-review rate is 47% even when the parcel arrives on time. The likely cause is the survey trigger — a customer who has received two of three shipments may be surveyed on an order that still feels incomplete.

This is a fulfilment or survey-timing issue, not a seller-quality issue, which is why it sits outside this analysis. It does mean the platform's true bad-review rate is **12.8%** rather than 12.35%, so every seller figure in this report is a conservative floor. **We recommend investigating it separately.**

2.3 Columns we created

From the cleaned base we derived the fields the analysis needs:

Column	How it was built	Why
is_bad	review_score ≤ 2	The outcome variable.
delay_days	actual delivery date – promised delivery date	Positive = late, negative = early.
is_late	delay_days > 0	Simple late/on-time split.
band	delay_days grouped into 6 bands, from "10+ days early" to "10+ days late"	A 1-day slip and a 3-week slip are different experiences. Banding also keeps the extreme tail without letting it distort averages.
gmV	Sum of item prices in the order	Lets us price every intervention in lost revenue.
category	Category of the highest-priced item, translated to English	One category per order for clean grouping.
bad_rate_ontime	Seller's bad-review rate counting only on-time parcels	The single most useful field in this report — it removes the delivery excuse.

3. Exploratory data analysis

Four charts establish the shape of the problem and identify the two factors we must control for before blaming any seller.

3.1 Review scores are U-shaped

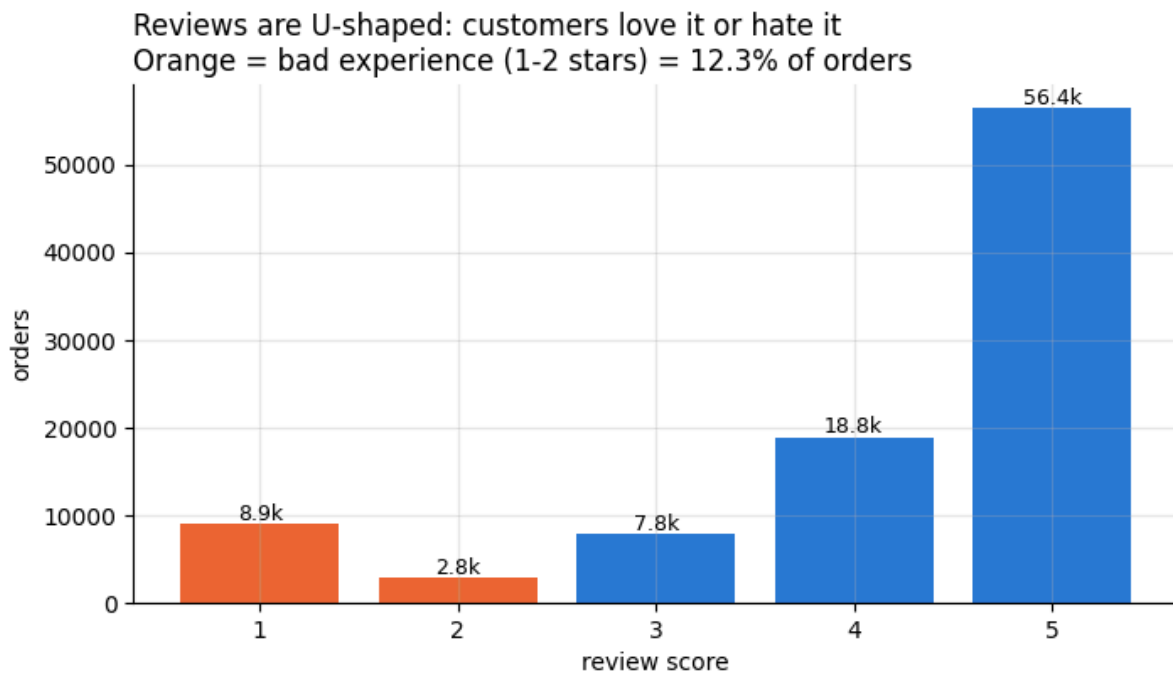


Figure 1. Distribution of review scores. Customers either love the order or hate it.

57% of reviews are five-star and 12% are one-star, with very little in between. This is why we measure the rate of 1–2 star reviews rather than the average score.

3.2 Late delivery is the strongest single driver

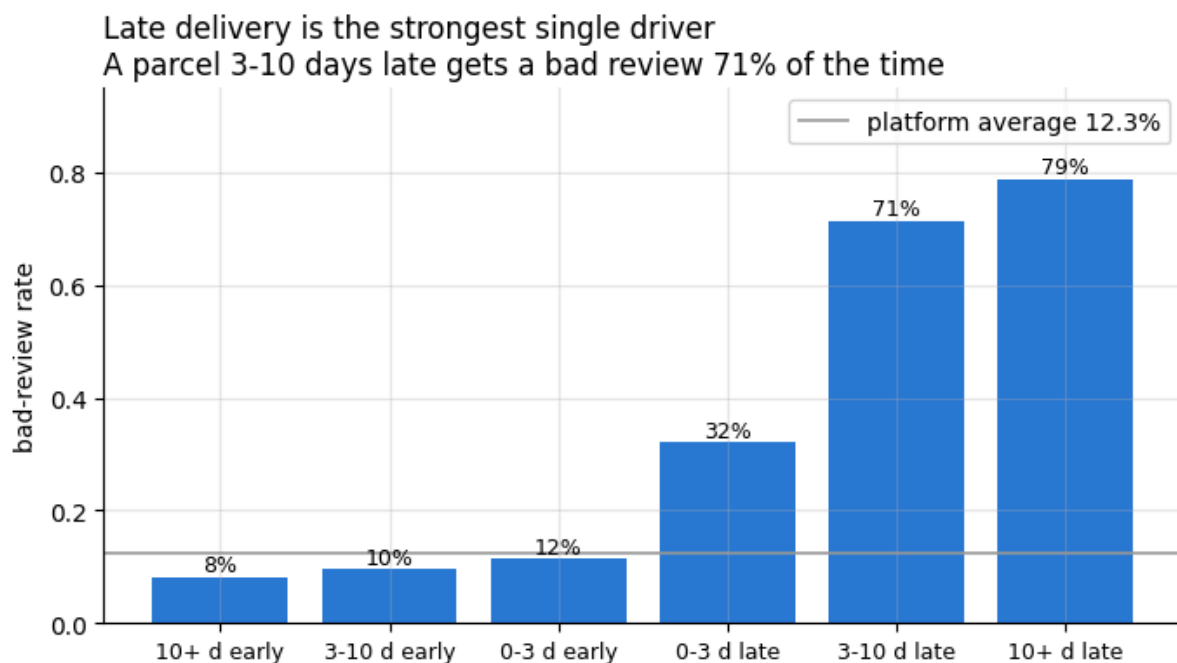


Figure 2. Bad-review rate by how the delivery compared with the promised date.

A parcel arriving 10+ days early draws a bad review 8% of the time. One arriving 3–10 days late draws one 71% of the time — a ninefold swing. This single chart is the reason seller bad-review rates cannot be compared directly: a seller shipping through a slow region would be condemned for its couriers.

3.3 Category matters, but much less

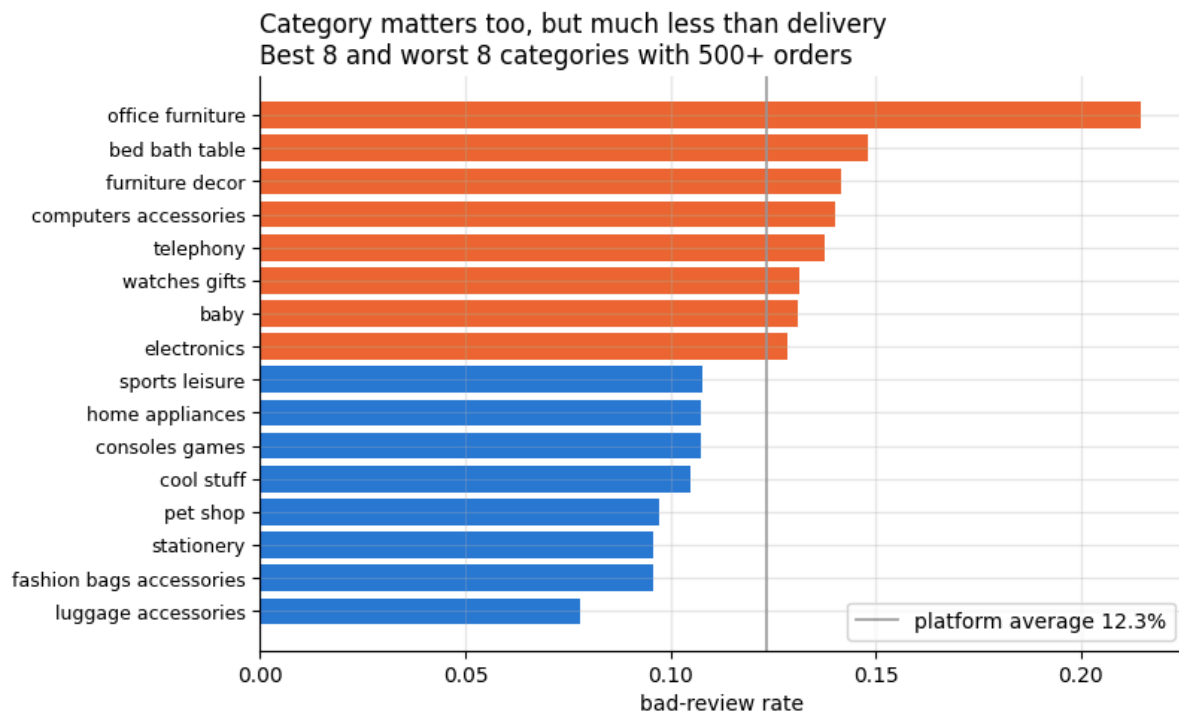


Figure 3. Best and worst eight product categories with 500+ orders.

Category moves the bad-review rate from about 8% to 21% — real, but far smaller than delivery. Office furniture is the worst category; luggage and fashion accessories the best.

3.4 The marketplace itself is highly concentrated



Figure 4. Cumulative share of orders as sellers are added largest first.

The largest 5% of sellers already handle 53% of all orders. This is essential context for Question 1: any statement of the form "the worst 5% of sellers cause X% of bad reviews" must be compared against the share of orders those sellers handle, or it simply rediscovers seller size.

WHAT THE EDA FORCES ON THE ANALYSIS

Because delivery timing and product category both move the bad-review rate substantially, we cannot judge a seller on its raw rate. Instead we give every order an **expected** bad-review rate predicted from its category and delivery band alone, then treat **actual minus expected** as the seller's own contribution.

The model reproduces the platform total to within **0.2%**, so the leftover is a trustworthy measure of seller responsibility rather than model error.

4. Question 1 — How concentrated is the damage?

4.1 Damage share compared with order share

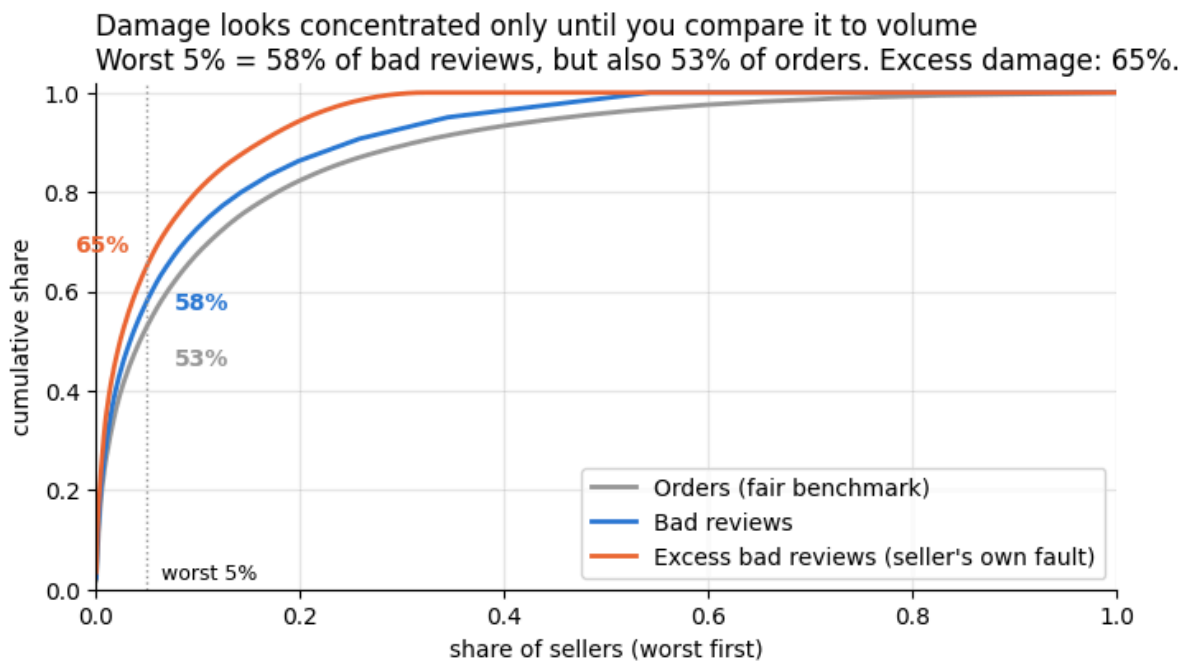


Figure 5. Cumulative share of damage as sellers are added worst-first, against the order-volume benchmark.

Worst x% of sellers	Share of bad reviews	Share of orders	Difference	Share of excess damage
1% (29 sellers)	29%	26%	+2.7 pp	35%
5% (147 sellers)	58%	53%	+5.1 pp	65%
10% (294 sellers)	73%	68%	+5.1 pp	80%
20% (588 sellers)	86%	82%	+4.0 pp	94%

The headline number is misleading. "The worst 5% of sellers cause 58% of bad reviews" is true, but those same sellers handle 53% of all orders — so only +5 percentage points of that is genuine concentration rather than size. Acting on this number would send the company after its own largest partners.

Concentration becomes real once we subtract what category and delivery already explain. On that measure the worst 5% own 65% of avoidable damage and the worst 1% own 35%.

4.2 "Worst 5%" means three different things

"Worst 5%" means three different things - only one is actionable

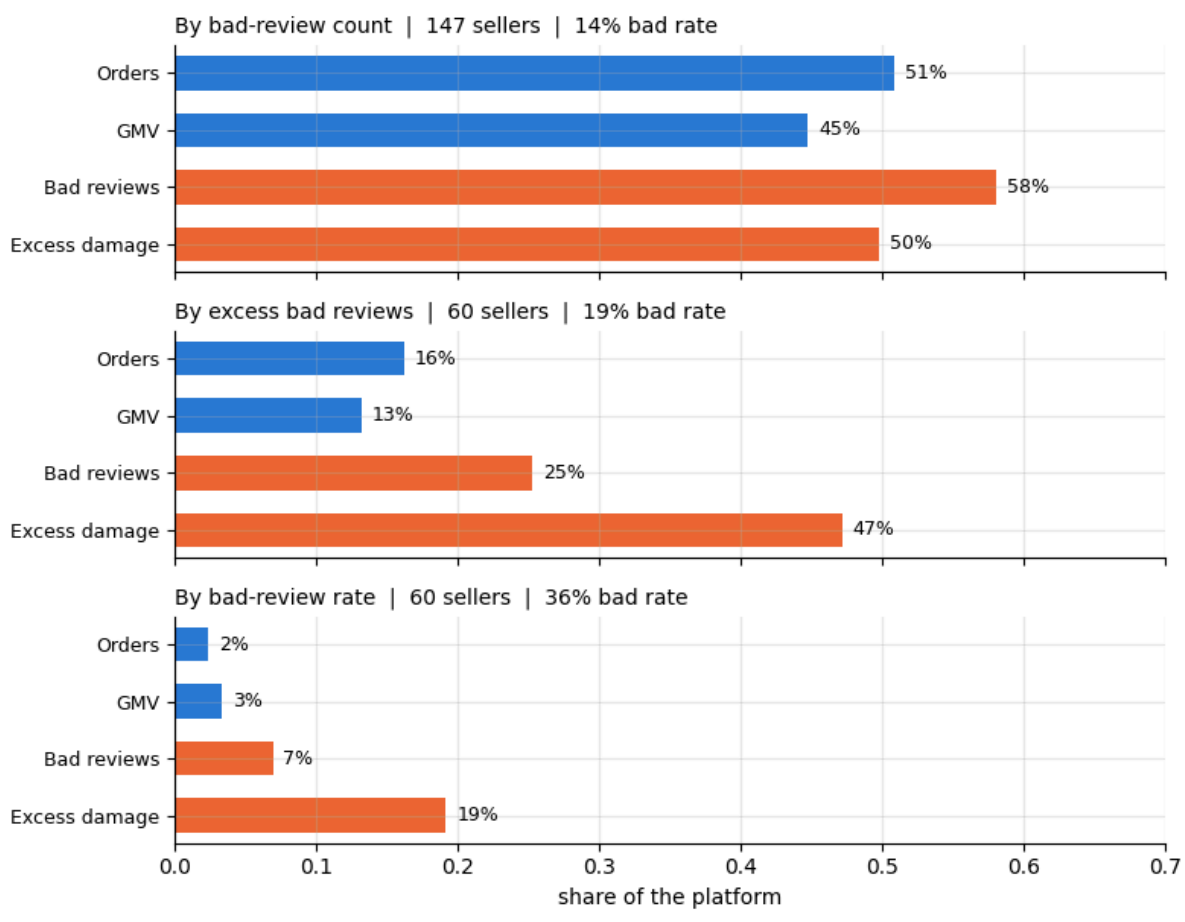


Figure 6. The same 5% cut under three definitions of "worst", showing what each cohort holds.

Ranked by	Sellers	Orders	GMV	Bad reviews	Excess damage	Their bad rate
Bad-review count	147	51%	45%	58%	50%	14%
Excess bad reviews	60	16%	13%	25%	47%	19%
Bad-review rate	60	2.4%	3.4%	7%	19%	36%

- Ranking by count picks the biggest sellers. Not a quality signal.
- Ranking by rate finds a genuinely toxic group (36% bad rate, three times the platform) but it handles only 2.4% of orders — too small to move the platform even if removed entirely.
- Ranking by excess damage finds the only cohort that is both large enough to matter and actually at fault: 13% of GMV and 47% of all avoidable bad reviews.

Less concentrated than the phrase suggests — and the honest answer depends entirely on how you rank sellers.

The worst 5% by bad-review count produce 58% of bad reviews but handle 53% of orders, so the genuine excess is only **+5 percentage points**. Almost all apparent concentration is the concentration of the marketplace itself.

Concentration is real in **excess** terms: the worst 5% own **65%** of the bad reviews that their categories and delivery records cannot explain. Total avoidable damage is **1,931 bad reviews — 17% of all of them**. That is the true size of the seller-side prize; the other 83% is delivery and category, not seller conduct.

5. Question 2 — Are certain sellers repeatedly bad?

We identified 77 sellers whose bad reviews are far above their own category-and-delivery expectation, then ran four tests, each designed to explain those sellers away.

The 77 sellers hold	Amount	Share
Orders	9,478	10.9% of qualified orders
GMV	R\$1.27M	10.9% of GMV
Bad reviews	2,264	20.6% of bad reviews
Excess (avoidable) damage	855	53.0% of avoidable damage

Ten percent of revenue is producing over half of all avoidable damage. The four tests below establish whether that is really their fault.

5.1 Test 1 — Is it just chance?

Threshold	Sellers observed	Expected by chance
3+ standard deviations above own expectation	33	1.6
4+ standard deviations	13	0.03

Chance alone would produce fewer than two sellers this extreme. There are 33. The distribution is also skewed +1.68, where pure random noise would be symmetric. This is not luck.

5.2 Test 2 — Is it the delivery?

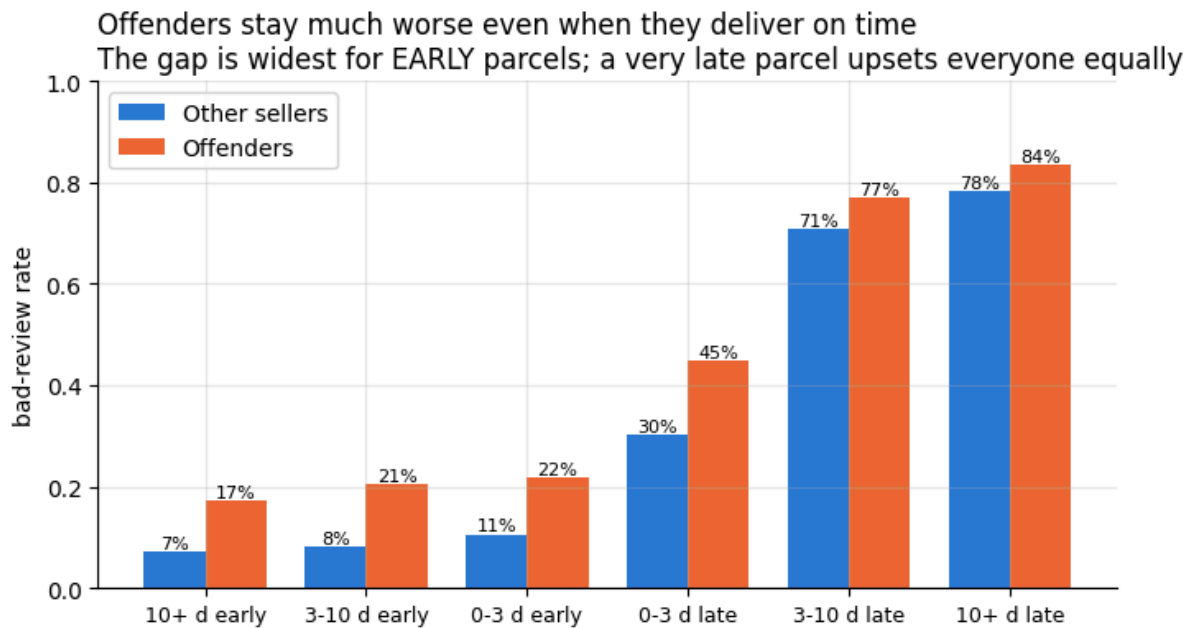


Figure 7. Bad-review rate by delivery band, offenders against all other sellers.

Counting only parcels that arrived on or before the promised date, the offenders draw a bad review 18.4% of the time against 7.6% for everyone else — 2.4 times worse with the logistics excuse removed. 77% of all their bad reviews are on parcels that were never late.

The shape of the gap is telling: it is widest where parcels arrived early, and almost disappears once a parcel is more than ten days late — at that point everyone leaves a bad review and seller identity stops mattering. Lateness does not distinguish these sellers; it hides them.

5.3 Test 3 — Is it the category?



Figure 8. Every seller's actual bad-review rate against what its category and delivery record predict.

The offenders' own categories predict a 12.7% bad-review rate. They deliver 28.2%. Of the 32 offenders selling in two or more categories, 41% are worse than the category baseline in every single category they trade in, and most of the rest in some. The problem travels with the seller, not the product line.

5.4 Test 4 — Was it just one bad patch?

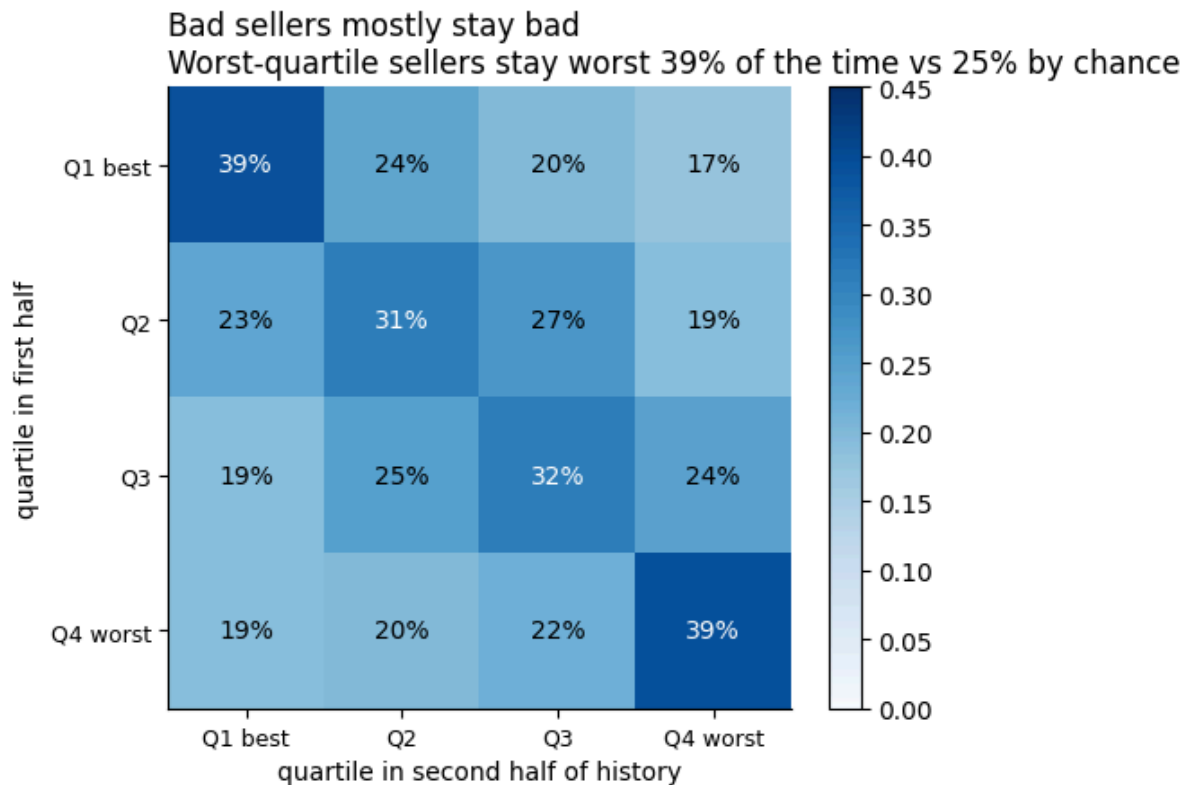


Figure 9. Where a seller lands in the second half of its history, given its first-half quartile.

Splitting each seller's order history in half chronologically, a seller in the worst quartile of the first half is still in the worst quartile 39% of the time, against 25% if it were random. Only 19% improve to the best quartile. The correlation between halves is 0.28 and highly significant.

ANSWER TO QUESTION 2

Yes — and every attempt to explain it away fails.

A group of **77 sellers** is repeatedly and measurably worse than its own category and delivery record predicts. They stay 2.4 times worse when their parcels arrive on time, sit 15 percentage points above their own categories' baselines, appear at extreme levels 20 times more often than chance allows, and stay in the worst quartile across time-split halves.

These sellers hold **11% of GMV** but **53% of all avoidable damage**.

6. What to do about it

6.1 Coach or delist?

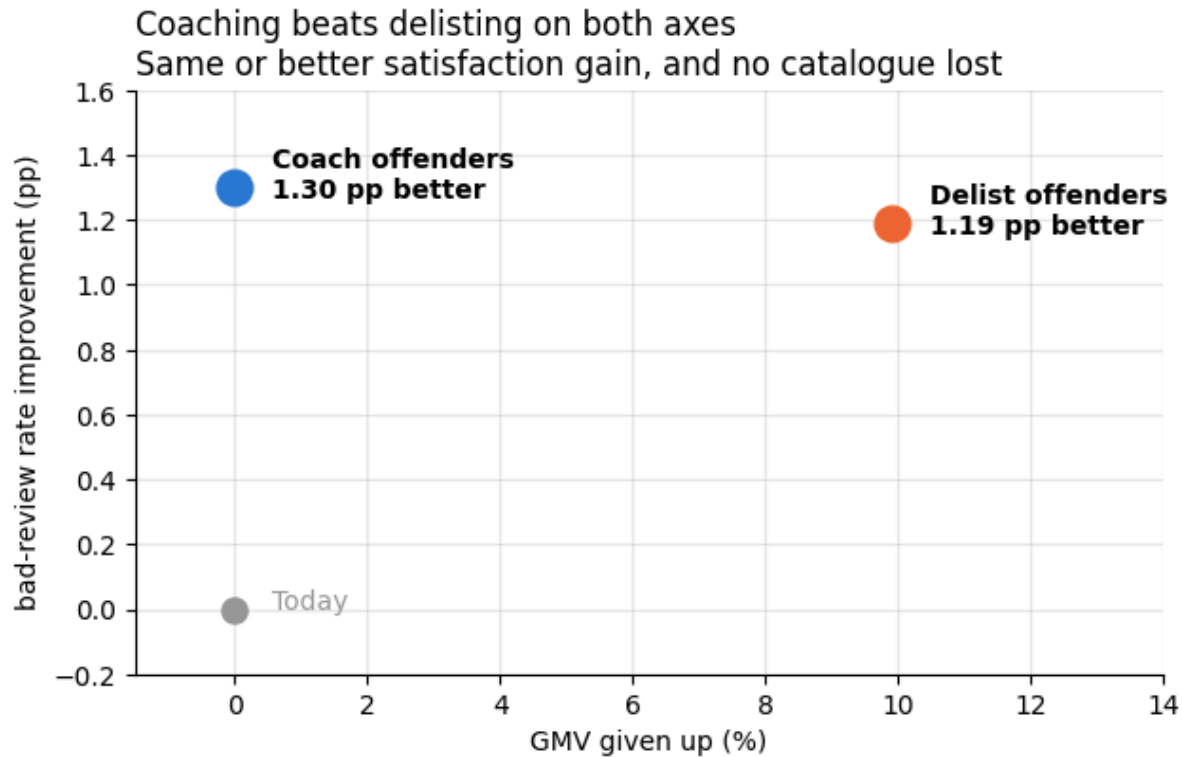


Figure 10. Improvement in the platform bad-review rate against the revenue each option costs.

Option	Platform bad rate	Change	GMV given up
Today — no action	12.35%	—	—
Coach the 77 sellers to the median	11.05%	−1.30 pp	None
Delist the 77 sellers	11.16%	−1.19 pp	9.9%

Coaching wins on both axes. This is counter-intuitive but arithmetic: delisting a seller removes its bad reviews, but also removes the roughly 78% of its orders that customers were perfectly happy with. The platform rate therefore improves less, and it costs a tenth of all revenue to achieve that smaller gain. Delisting should be the consequence of a failed coaching window, not the opening move.

6.2 The 77 sellers are not one problem

Failure mode	Sellers	Bad rate	Bad rate when on time	Late rate	Recommended action
Borderline — watch	37	21%	18%	5.6%	Monitor one more quarter
Product — item disappoints	26	33%	32%	3.7%	Listing audit and quality hold
Mixed — both problems	9	43%	36%	19%	Throttle volume until both clear
Fulfilment — ships slowly	5	31%	21%	16%	Shipping SLA and ops coaching

The Product group is the important case. It ships late only 3.7% of the time — better than the platform average of 8.1% — and is still disliked 32% of the time. No delivery SLA would reach these sellers; their problem is that the item is not what the listing promised. Treating all 77 as a logistics problem would miss most of the damage.

6.3 Recommendations

1. Track excess bad reviews, not raw complaint counts. Counts point at your biggest sellers; excess points at the ones actually at fault.
2. Coach the 77 sellers rather than delisting them: a larger satisfaction gain, and no revenue lost.
3. Split the programme by failure mode. Fulfilment cases need a shipping SLA; product cases need a listing and quality review, because an SLA would do nothing for them.
4. Add the on-time bad-review rate to the seller scorecard. It is the most diagnostic field in this analysis, it separates a courier problem from a product problem, and it needs no model at all.
5. Intervene early. Because seller quality is persistent, a seller whose first ten orders are already far above expectation is unlikely to correct itself. Act at order ten, not order two hundred.
6. Recruit alternative office-furniture supply before enforcing there. That category has the worst bad-review rate and most of its volume sits with sellers on this list, so enforcement without recruitment would remove much of the category.
7. Do not expect seller action alone to fix satisfaction. The entire seller lever is worth about 1.3 points of the 12.35% bad-review rate. Delivery timing is the larger opportunity.

7. Limitations

- Multi-seller orders (1.3%) were excluded because one review cannot be split between sellers. Those orders have a 47% bad-review rate, so the platform's true rate is 12.8% and all seller figures here are conservative. This effect deserves its own investigation.
- Reviews are a proxy for satisfaction, not a census. Coverage is high (99% of orders have a review), but a customer who leaves quietly never appears in the data.
- The expected-rate model controls for product category and delivery band only. Price tier, product weight, freight cost and seasonality are not controlled, so some residual damage may reflect those rather than seller conduct.
- Sellers with few orders have noisy rates. This is why the list requires 10+ orders and a statistical threshold rather than a raw rate — but any individual small seller on the list should be reviewed manually before action.
- The coaching estimate is a "what if they matched the median seller" calculation, not a measured result. It is a ceiling on what is available, not a forecast.
- The data covers two years and one marketplace. Nothing here has been validated out of sample.

8. Reproducing this analysis

All code is in the companion notebook, which runs top to bottom from the raw CSV files and regenerates every figure in this report:

[Seller_Performance_Analysis.ipynb](#)

It requires pandas, numpy, scipy and matplotlib, and should be run from the project folder — the one containing the "E-Commerce Dataset" directory. It writes its charts to a figures/ folder and the full 77-seller list to seller_watchlist.csv.

Python and pandas were chosen because the work is relational joins plus group-wise aggregation, where a dataframe library is the shortest reliable path. Matplotlib was used directly rather than through a higher-level charting library because several figures are non-standard forms — a damage curve benchmarked against a second curve, an actual-versus-expected scatter with a reference diagonal, a quartile transition matrix. Chart colours were checked for colour-blind readability, and no chart in this report uses two vertical axes or a value label on every point.